Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/03/2020
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Space Flight Environment Induces Remodeling of Vascular Network and Glia-Vascular Communication in Mouse Retina		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS Rodent Research-9 NOTE: End date changed to 10/31/2021 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: End date changed to 3/31/2020 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 4/30/2019 per F. Hernandez/ARC (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University) Sun, Shu-Wei Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX15AB41G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The eye is a unique organ because it is relatively unprotected and constantly exposed to atmospheric oxygen, background radiation, environmental chemicals, and physical abrasion. A recent report shows that more than 30% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. Our Preliminary studies from mice that had been subjected to space flights showed that environmental conditions during space travel lead to oxidative stress and induce adverse microvessel remodeling in the retina. However, our knowledge about the pathophysiological process from adaptive response to irreversible oxidative damage in the retina during and after space flight exposure and the underlying cellular mechanism(s) of these disturbances is very limited and inconsistent. The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. The hypothesis will be tested in the specific aims with the space flight experiments on the International Space Station (ISS) and compared to that of ground-based groups using matured female mice. For flight studies, there will be two experiments over two flights, one for in-flight adaptation study on the ISS and one for studying re-adaptation after live animal return to Earth following long-term ISS mission. Selected groups of mice will be treated intraperitoneally with an antioxidant a week before launch and weekly during space flight. Animals will be sacrificed in orbit at 14 and 30 days after launch or 30 days and up to 6 months after return alive on Earth. Together, our unique, integrative, and quantitative activities with advanced MRI (magnetic resonance imaging) imaging techniques and retinal functional testing using visual electroretinography will provide insight into the molecular mechanisms and pathways of space flight condition-induced mitochondrial oxidative damage on retinal BRB integrity, vascular remodeling, and retinal function. The data will provide unique criteria for risks of functional detriments associated with space flight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our proposed study from space flight mission to ISS will provide a unique opportunity to evaluate the effects of space hazards and mechanisms of response and remodeling of the ocular system. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Task Progress:	SpaceX-12 was launched at the Kennedy Space Center (KSC) on a 30-day mission on August 14, 2017. Nine week old male C57BL/6 mice (n=20) were flown on SpaceX-12 in NASA's animal enclosure modules (AEMs). The animals then were returned to Loma Linda University (LLU) where they were sacrificed followed by dissections. Experiment with ground-based control groups have been completed in May, 2018. Fixed ocular samples have been used for histochemical analysis for identifying biomarkers for oxidative stress, apoptosis, and blood-retina-barrier (BRB) function. Frozen retina samples are used for genomics, proteomics, and metabolomics analysis. Our data showed that flight group had significant apoptosis in the retina and retinal vascular endothelial cells compared to control groups ($p<0.05$). Immunohistochemical analysis of the retina revealed that an increased expression of aquaporin-4 (AQP-4) in the flight mice compared to controls gave strong indication of disturbance of BRB integrity. There was also a significant increase in the expression of platelet endothelial cell adhesion molecule-1 (PECAM-1) and a decrease in the expression of the BRB-related tight junction protein, Zonula occludens-1 (ZO-1). Proteomic analysis showed that many key proteins and pathways responsible for cell death, cell cycle, immune response, mitochondrial function, and metabolic stress were significantly altered in the flight mice compared to ground control animals. These data indicate a complex cellular response that may alter retina structure and BRB integrity following long-term space flight. Our study also showed that many proteins and pathways responsible for mitochondrial function, nervous system development, behavior, protein/organelle transport, and metabolism were significantly altered after space flight in the brain. Pathways associated with adhesion molecular remodeling were also significantly altered. Our data demonstrate that exposure to the space flight environment induces significant changes in protein expression related to neuronal structure, neurovascular integrity, and metabolic function. The micro-CT analysis showed that the cross-section areas of the retina, retinal pigment epithelium (RPE), and choroid layer thickness were changed in space flight samples compared to ground controls (GC), with space flight samples showing significantly thinner cross-sections and layers compared to controls. The findings from this study indicate that micro-CT evaluation is a sensitive and reliable method to characterize ocular structure changes. These results are expected to improve the understanding of the impact of environmental stress on global ocular structures.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XW, Nishiyama NC, Stanbouly S, Holley J, Britten R. "Prolonged exposure of low-dose neutron irradiation impacts on rat blood-retina barrier integrity and ocular structure." Presented at 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
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Articles in Peer-reviewed Journals	Mao XW, Nishiyama NC, Byrum SD, Stanbouly S, Jones T, Holley J, Sridharan V, Boerma M, Tackett AJ, Willey JS, Pecaut MJ, Delp MD. "Spaceflight induces oxidative damage to blood-brain barrier integrity in a mouse model." FASEB J. 2020 Nov;34(11):15516-30. Epub 2020 Sep 26. https://doi.org/10.1096/fj.202001754R ; PMID: 32981077 , Nov-2020